

NAME:

SOLUTIONS

Math 553 — Spring 2026— Midterm 1

Total points 100.

Questions: 4

Explain all answers. If you cannot solve a problem, at least explain the relevant theorems or methods.

Duration: 1 hour (you may stay for up to 2 hours)

Books and notes may be used (anything on paper).

No communication or consultation with any person or electronic resource about the exam.

No laptops, electronic devices or phones, except that you may consult books and notes you have stored on these devices. Web searching and LLMs are not permitted.

Open note files on your devices BEFORE the start of the exam. No typing during the exam.

Writing on the back of the page will not be graded.

Put any extra work on the last page (blank).

Honor code statement

I did not use electronic devices, except to consult books and notes on my devices. I did not communicate or consult with any person or electronic resource about the contents of this exam.

Signature:

Date: Thursday March 5, 2026

Problem 1. (30 points) Let $G(z) = z^2 - z$. (Not Burgers, not traffic flow.)

Solve the conservation law $G(u)_x + u_y = 0$ for $x \in \mathbb{R}, y > 0$, with initial data

$$u(x, 0) = \begin{cases} 1/2 & \text{for } x > 0, \\ 0 & \text{for } x < 0. \end{cases}$$

Include a well-labelled sketch of the projected characteristics. If there is a shock, check the entropy condition.

If there is a fan, find the formula for $u(x, y)$ in the fan. You may use results from class.

Solution:

See next page.

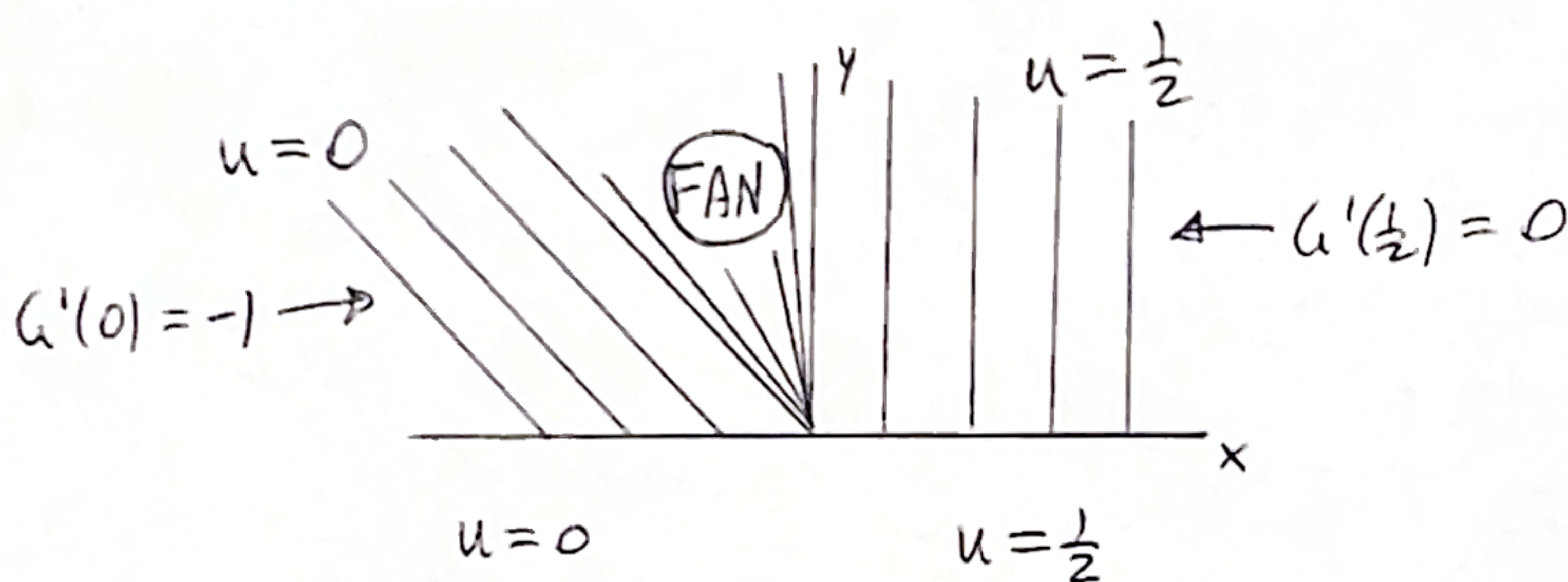
Problem 1.

$$G'(z) = 2z - 1$$

$$\begin{cases} \frac{dx}{dt} = G'(z) & \Rightarrow x = G'(z)t + c_3 \\ \frac{dy}{dt} = 1 & \Rightarrow y = t + c_1 \\ \frac{dz}{dt} = 0 & \Rightarrow z = c_2 = \text{constant, along the projected characteristics.} \end{cases}$$

$$\Rightarrow x = G'(z)y + c_4$$

Note $G'(0) = -1$, $G'(\frac{1}{2}) = 0$.



In the fan region we guess $u = v(x/y)$ for some function v . Then

$$G'(u)u_x + u_y = 0$$

$$\Rightarrow (2v - 1)v'(x/y) \frac{1}{y} + v'(x/y) (-x/y^2) = 0$$

$$\Rightarrow 2v - 1 = x/y$$

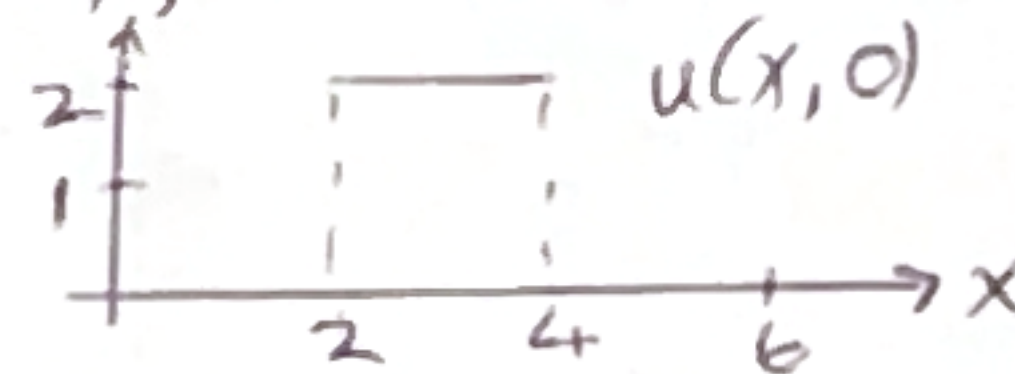
$$u = v(x/y) = \frac{x/y + 1}{2} \quad \text{in the fan, which is}$$

constant on projected characteristics $x = (\text{const.})y$.

Note $v(x/y) = \frac{1}{2}$ on the line $x=0$, and $v(x/y) = 0$ on $x=-y$.

Problem 2. (20=3+7+7+3 points) Consider the wave equation with $c = 1$, on the interval $0 < x < 6$:

$$\begin{cases} u_{tt} - 1^2 u_{xx} = 0 \\ u(x, 0) = 2 \text{ for } 2 < x < 4; 0 \text{ otherwise} \\ u_t(x, 0) = 0 \\ u(0, t) = 0 \\ u_x(6, t) = 0 \end{cases}$$



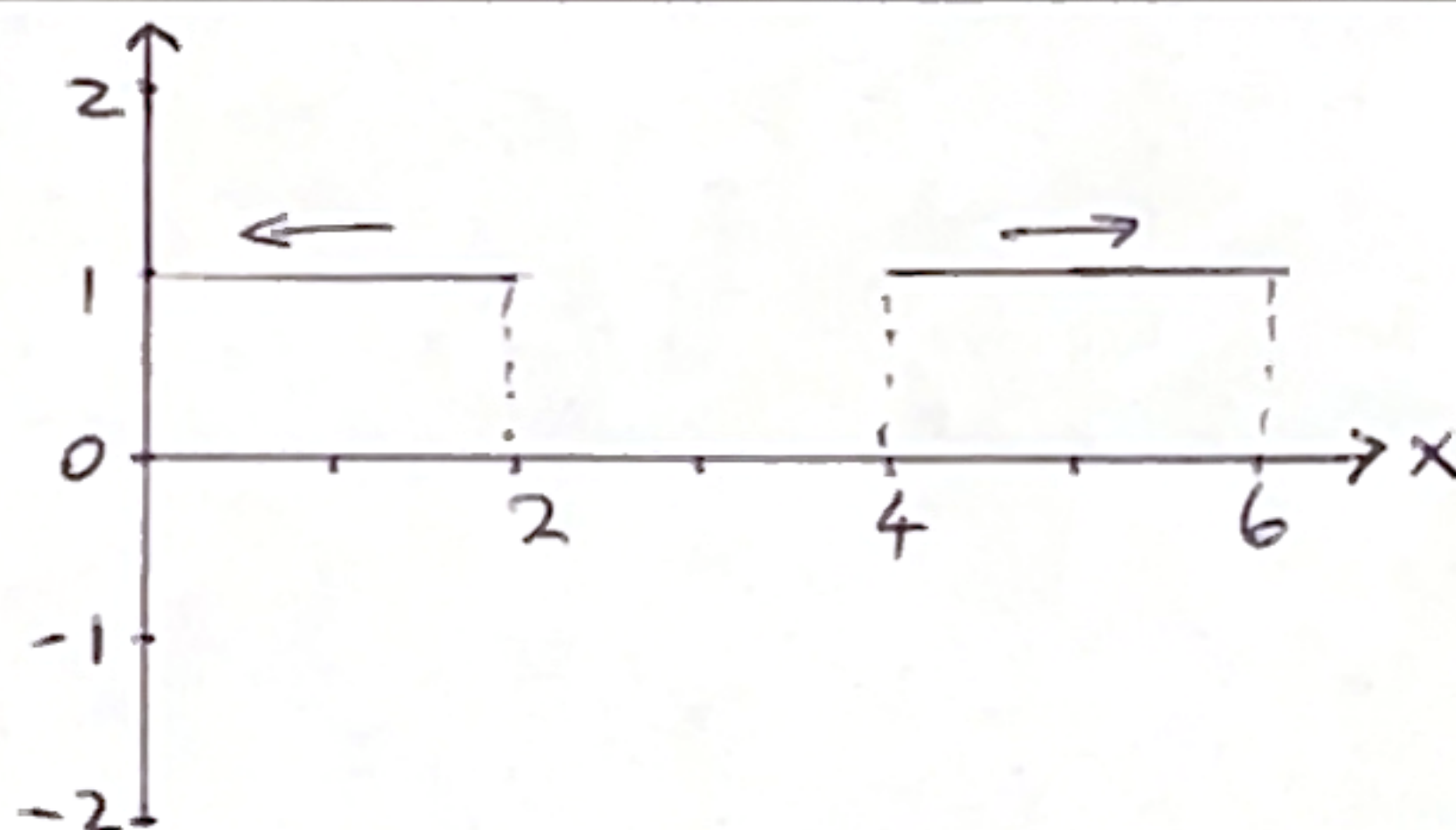
(a) What type of boundary condition is at $x = 0$? and at $x = 6$?

Solution: $x = 0$: Dirichlet $x = 6$: Neumann

(b) Sketch the solution at time $t = 2$.

Solution:

$t = 2$:

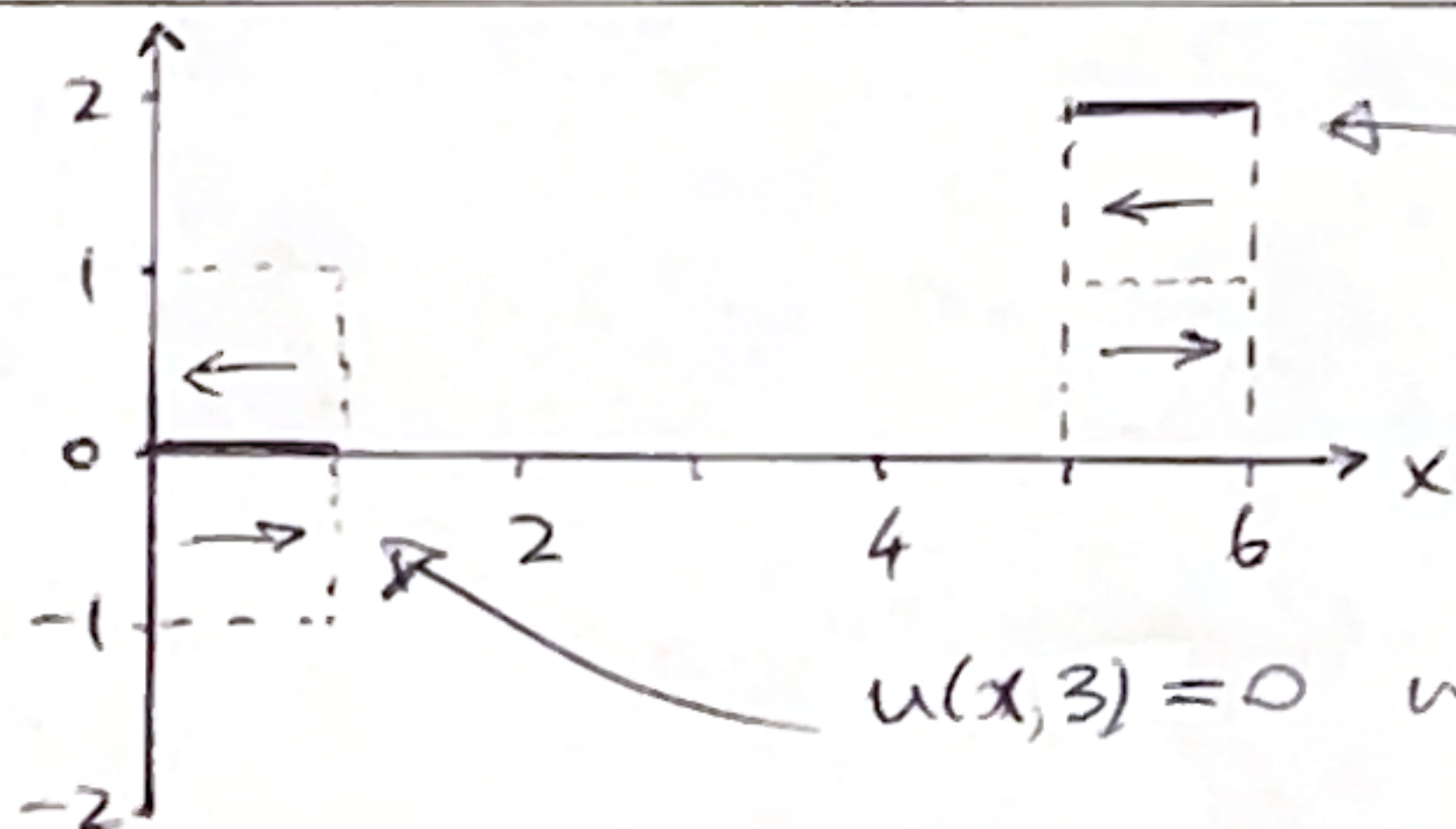


Release from rest ($u_t(x, 0) = 0$). So D'Alembert says half the wave goes left, half right, speed $c = 1$.

(c) Sketch the solution at time $t = 3$.

Solution:

$t = 3$:



$u(x, 3) = 2$ when $5 < x < 6$

$u(x, 3) = 0$ when $0 < x < 1$

(d) Briefly explain your reasoning for (b) and (c). Do not use formulas!

Solution: At $x = 0$ (Dirichlet) the wave reflects upside down and so cancels completely, at $t = 3$. At $x = 6$ (Neumann) the wave reflects the same way up and so adding the original wave and reflected wave gives height 2, at $t = 3$.

Problem 3. (25=20+5 points) Consider the nonhomogeneous wave equation $u_{tt} - c^2 \Delta u = f$ on $\mathbb{R}^n$. Let

$$E(t) = \frac{1}{2} \int_{\mathbb{R}^n} (u_t^2 + c^2 |\nabla u|^2) dx.$$

Assume u and f are smooth and have compact support, at each time t .

(a) Show $E'(t) = \int_{\mathbb{R}^n} u_t f dx$.

Solution:

$$\begin{aligned} E'(t) &= \frac{d}{dt} \frac{1}{2} \int_{\mathbb{R}^n} (u_t^2 + c^2 \nabla u \cdot \nabla u) dx \\ &= \int_{\mathbb{R}^n} (u_t u_{tt} + c^2 \nabla u \cdot \nabla u_t) dx \quad (\text{differentiation through the integral is valid since } u_t \text{ and } \nabla u \text{ have compact support}) \\ &= \int_{\mathbb{R}^n} (u_t [c^2 \Delta u + f] + c^2 \nabla u \cdot \nabla u_t) dx \quad \text{by the PDE for } u_{tt} \\ &= \int_{\mathbb{R}^n} (u_t f + c^2 \nabla \cdot [u_t \nabla u]) dx \\ &= \int_{\mathbb{R}^n} u_t f dx + c^2 \int_{\{\text{boundary of large ball}\}} [u_t \nabla u] \cdot \nu dS(x) \quad \text{by the divergence theorem} \\ &= \int_{\mathbb{R}^n} u_t f dx \quad \text{since } \nabla u \equiv 0 \text{ on the boundary of the large ball, by compact support.} \end{aligned}$$

Comment: an equivalent way to organize this calculation is to multiply the PDE by u_t and integrate with respect to $x \in \mathbb{R}^n$.

(b) Explain why the formula $E'(t) = \int_{\mathbb{R}^n} u_t f dx$ is intuitively reasonable, for example if $f > 0$.

(You may take $n = 1$, so that f represents an external force on the vibrating string.)

Solution: If $f > 0$ and $u_t > 0$ then the force on the string and the velocity of the string are both upward, so that the force makes the velocity more positive, and so increases the kinetic energy above what it would have been otherwise. Apart from the effect of the external force, energy is conserved, and so one now expects $E'(t) > 0$, which is consistent with the formula.

If $f > 0$ and $u_t < 0$ then the force on the string is upward and the velocity of the string is downward. In this case, the force will make the velocity less negative, hence decreasing its square and so decreasing the kinetic energy below what it would have been otherwise. Thus one expects $E'(t) < 0$, which is consistent with the formula.

Comment: when you are asked for intuition about a formula in a mathematics course, the first thing you check is whether the sign is correct. That is why we focus on whether $E'(t)$ is positive or negative, in the two cases.

Problem 4. (25=20+5 points) Assume u solves the wave equation in 3 dimensions and is radially symmetric, meaning $u(r\xi, t)$ depends on r and t but not on the direction of the unit vector ξ .

(a) Show that for some functions F and G , we have

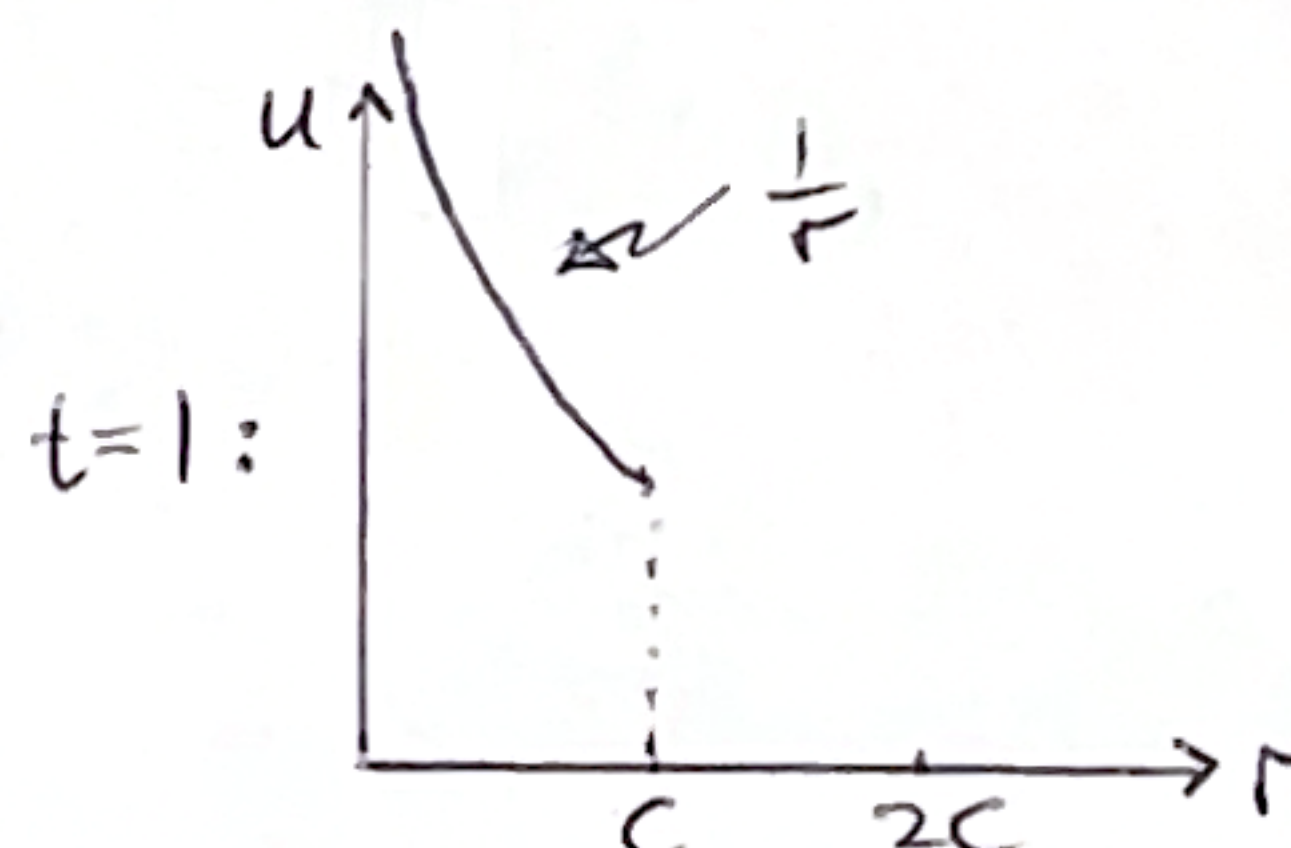
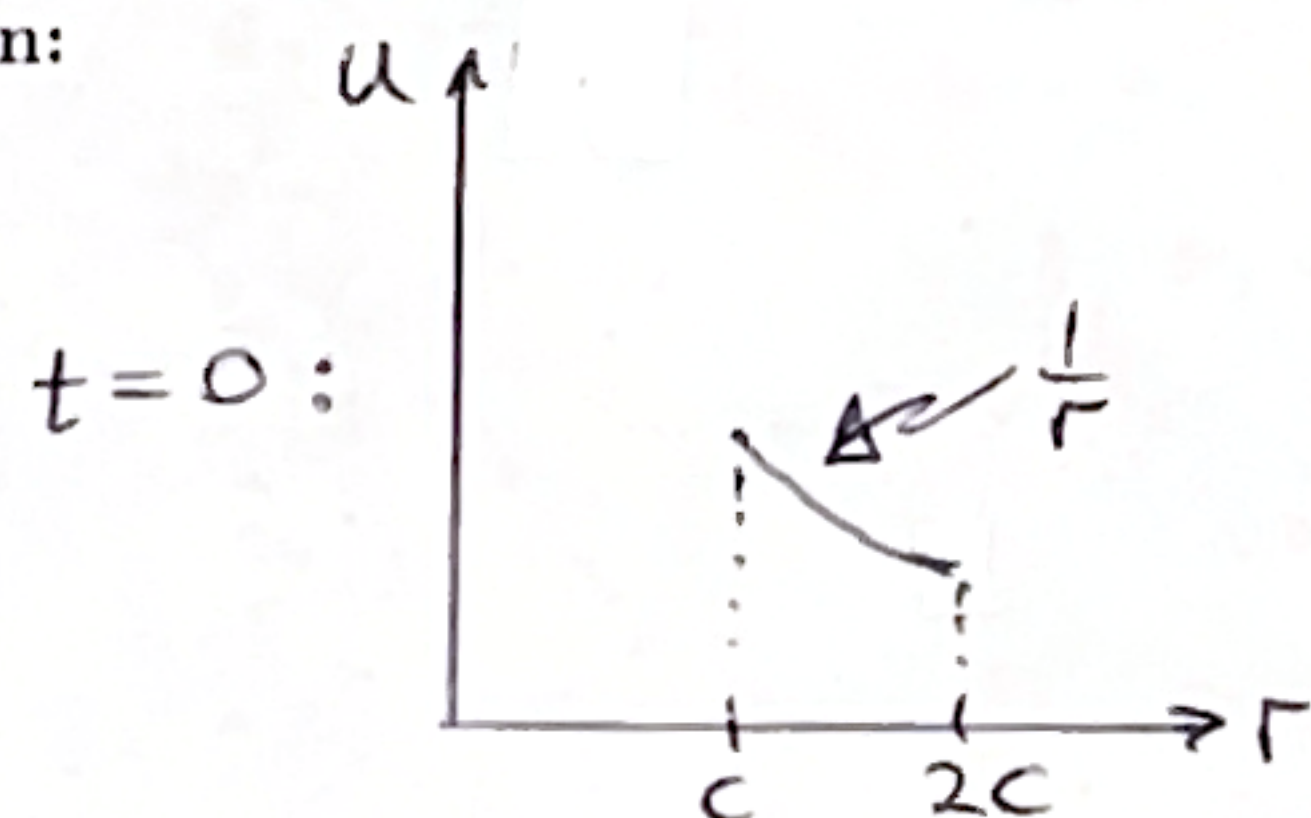
$$u(r\xi, t) = \frac{F(r+ct) + G(r-ct)}{r}.$$

Hints. Show $u(r\xi, t) = M_u(0, r)$, by radial symmetry. From class, $rM_u(0, r)$ satisfies a wave equation.

Solution: $M_u(0, r) = \int_{S^2} u(0+r\xi, t) \frac{d\xi}{4\pi} = u(r\xi, t) \cdot \int_{S^2} \frac{d\xi}{4\pi} \rightarrow 1$
 $\Rightarrow M_u(0, r) = u(r\xi, t)$ } since u is radial so does not depend on ξ
 Next, from class we know $rM_u(0, r)$ satisfies the 1-dim. wave eq. (see the proof of Kirchhoff) with respect to the r and t variables.
 So by trav. wave soln:
 $r u(r\xi, t) = r M_u(0, r) = F(r+ct) + G(r-ct)$
 for some functions F and G .
 Now divide through by r .

(b) Let $F(s) = \begin{cases} 1, & c < s < 2c \\ 0, & \text{otherwise} \end{cases}$ and $G \equiv 0$. Sketch snapshots of $u(x, t)$ at $t = 0$ and at $t = 1$.

Solution:



(c) (No credit, just for interest) "An inward-moving spherical wave focuses into a point singularity." Discuss.

Solution: $F(r+ct)$ moves to the left and so $F(r+ct)/r$ represents an inward moving wave.
 At $t=1$, it has created an infinite height at $r=0$:
 that is, a singularity! (i.e. toward smaller r -values)